Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/09/2025
PI Name:	Turek, Fred W Ph.D.		
Project Title:	Effects of Spaceflight on Gastrointestinal Microbiota in Mice: Mechanisms and Impact on Multi-System Physiology		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	fturek@northwestern.edu	**Fax:**	FY 847-467-4065
PI Organization Type:	UNIVERSITY	**Phone:**	847-467-6512
Organization Name:	Northwestern University		
PI Address 1:	Center for Sleep and Circadian Biology		
PI Address 2:	2205 Tech Drive, Hogan Hall 2-160		
PI Web Page:	http://www.northwestern.edu/		
City:	Evanston	**State:**	IL
Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 5/31/2026 per F. Hernandez/ARC (Ed., 7/1/25) NOTE: End date changed to 5/31/2025 per F. Hernandez/ARC (Ed., 10/22/24) NOTE: End date changed to 9/30/2024 per F. Hernandez/ARC (Ed., 6/14/24) NOTE: End date changed to 5/31/2024 per NSSC information (Ed., 3/17/24) NOTE: End date changed to 5/31/2023 per NSSC information (Ed., 6/15/22) NOTE: End date changed to 5/31/2022 per NSSC information (Ed., 9/23/21) NOTE: End date changed to 5/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 7/17/20) NOTE: End date changed to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/14/19)		

	NOTE: End date changed to 5/31/2019 per NSSC information (Ed., 6/7/18)
Key Personnel Changes/Previous PI:	June 2021 report: Co-Investigator Stefan Green has moved to Rush University from University of Illinois, Chicago (UIC). 2025 report: Dr. Peng Jiang has now taken a position at Merck.
COI Name (Institution):	Green, Stefan Ph.D. (Rush University) Keshavarzian, Ali M.D. (Rush University Medical Center) Vitaterna, Martha Ph.D. (Northwestern University) Jiang, Peng Ph.D. (Northwestern University) Summa, Keith (Northwestern University)
Grant/Contract No.:	NNX15AL05G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal in response to NNH14ZTT002N NASA Research Announcement (NRA) “Research Opportunities for Flight Experiments in Space Biology” is intended to examine the potential role(s) of disruption of microbial communities in the gut (dysbiosis) in mammalian adjustment to the space environment. Many challenges to mammalian physiological homeostasis present in the space environment are known to create dysbiosis. Similarly, immune or inflammatory changes, as well as metabolic alterations, observed with time in spaceflight may be attributable in part to dysbiosis. A series of experiments in mice will be carried out to define the mechanisms by which time in space, diet, and host genotype interact to impact the composition of the gut microbiota, and how dysbiosis relates to gene expression (by RNA-seq) and physiology in serum, colon, ileum, spleen, liver, and fat as well as the sleep/wake cycle and feeding behavior. Extensive use of ground based studies will support the analysis and interpretation of spaceflight data. Five projects at three institutions will focus on different aspects. Project 1 (University of Illinois at Chicago, Principal Investigator (PI: SJ Green) will carry out metagenomic sequencing of microbiota from fecal samples and cecal contents (including from the tissue sharing opportunity); Projects 2 and 3 (Rush University Medical Center, PIs: A Keshavarzian and C Forsyth) will use gene expression, physiological and histologic measures of gastrointestinal and immune function, respectively; while Projects 4 and 5 (Northwestern University, PIs: FW Turek and MH Vitaterna) will focus on metabolism, sleep, and circadian behavioral organization. Dr. Turek (Northwestern) will provide Principal Investigator (PI) oversight. These studies are in alignment with NASA Research Announcement Research Emphases 1)“Spaceflight Omics Studies” and 3)“Understanding How Complex Organisms Adapt to Spaceflight.”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal will examine the impact of the space environment on the population structure of the intestinal microbiota of mice, and how multiple physiological systems involving the gastrointestinal (GI), immune, metabolic, circadian, and sleep systems, known to be affected by the microbiota, are impacted by the space environment. While our overall working hypothesis is that the space environment and associated stressors will have a negative impact on the microbiota (i.e., dysbiosis), and that this in turn will negatively impact multiple physiological systems known to be adversely affected by spaceflight, our integrated analysis of gene expression and multiple tissue/organ specific biomarkers of tissue/organ injury will also allow us to uncover the adverse effects of spaceflight due or not due to dysbiosis. However, the impact of the proposed studies extends beyond understanding the mechanisms at play in the unique stresses of spaceflight. The proposed studies will elucidate mechanisms underlying interactions between GI, immune, metabolic, and sleep functions and specifically the role of the microbiota in these interactions. This knowledge has tremendous potential for guiding development of dysbiosis-targeted interventions for disorders in all of these systems.
Task Progress:	Metagenomic sequencing microbiome analysis using STARMAPs (Jiang et al., Microbiome, 2019) focuses on the changes or differences in microbiome composition between groups rather than the composition of the microbiome itself. [Ed. Note: "Similarity Test for Accordant and Reproducible Microbiome Abundance Patterns" (STARMAP). See References.] This approach is extremely informative for comparisons between experiments, since the composition can differ greatly between cohorts of animals, even from the same source. Using this approach, we have identified flight vs. ground differences reproducible between NASA's Space Transportation System-135 (STS-135), Rodent Research-1 (RR-1), and Rodent Research-7 (RR-7) missions. We are expanding our analysis to additional spaceflight missions, as well as exploring new data sets characterizing the impacts of Earth-based stressors that have been hypothesized to contribute to, or mimic, the effects of spaceflight. We also have found significant similarity in the effects of rapid eye movement (REM) sleep disruption and spaceflight, and identified a prebiotic fiber supplementation that has opposite effect on microbiome composition (i.e., a potential countermeasure). Having modified the code for STARMAPs to encompass use of a greater range of data sets, the analyses of aging, cold exposure, intermittent hypercapnia, and other data sets are underway (Kim, SY, et al., manuscript in preparation). Metagenomic sequencing of the ground control re-run (Blue-enriched vs. White light study) has been completed; fecal samples corresponding to the same timepoints and conditions as the original study were examined. This was done under the supervision of co-PI Stefan Green (Director of Genomics and Microbiome Core Facility, Rush University), using primers and methods comparable to the original study. Overall, the results highlight a lack of any significant effect of the different light conditions on the mice gut microbiomes. Minor effects seen in the Permutational Multivariate Analysis of Variance (PERMANOVA) model (and to some extent in the principal component analysis/PCA plots) are most likely attributable to a “founder effect”, as the effect was present even at the baseline. From this preliminary analysis, we feel confident in moving forward with analyzing the RR-7 datasets, and our concerns are allayed regarding the mismatched light spectrum affecting the microbiome. REFERENCES Jiang P, Green SJ, Chlipala GE, Turek FW, Vitaterna MH. "Reproducible changes in the gut microbiome suggest a shift in microbial and host metabolism during spaceflight." Microbiome. 2019 Aug 9;7(1):113. https:// ; PMID: 31399081; PMCID: PMC6689164
Bibliography Type:	Description: (Last Updated: 07/16/2025)

Conference Materials (Downloadable)	Vitaterna MH, Campbell KJ, Jiang P, Lin X, Olker C, Lewiston J, Song EJ, Choi S, Turek FW. "Mouse strain differences in activity/rest and sleep/wake behaviors in spaceflight and ground-based analog stress." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024 40th Annual Meeting for the American Society for Gravitational and Space Research.pdf